



Securing understanding of rounding

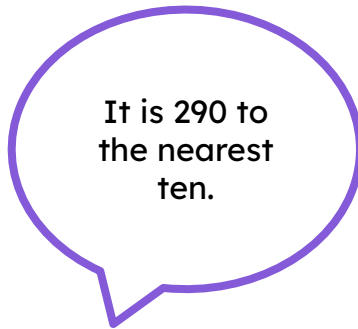
Task A: Unrounded numbers

1) Sofia is thinking of some numbers. She uses rounding to give two clues. After each clue, guess her number and write it down. Does your guess change each time?

a) Clue 1



Clue 2

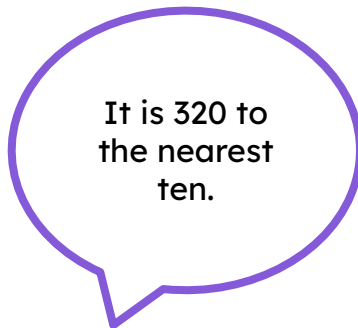


E.g. 282
292
Yes

b) Clue 1



Clue 2

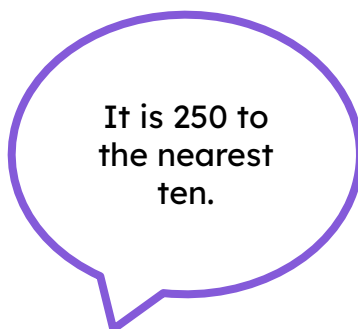


E.g. 329
317
Yes

c) Clue 1



Clue 2



E.g. 254
254
No

2) Jun's homework has been damaged by his dog!

His answers are still there but some of the numbers are missing from the questions. What could the missing numbers be?

- a) 331 to the nearest *hundred* is 300
- b) 478 to the nearest *thousand* is 0
- c) 1785 to the nearest hundred is 1800
- d) 4995 to the nearest *thousand* is 5000
- e) 397 to the nearest ten is 400

3) a) Are these statements always, sometimes or never true?

A number rounded to the nearest 10 will have two digits.

A number rounded to the nearest 10 will end in one zero.

A number rounded to the nearest 10 will be less than 5 away from the original number.

All statements are sometimes true.

b) Give some examples to show why.

E.g. 192 to the nearest 10 is 190 and has 3 digits.

E.g. 1995 to the nearest 10 is 2000 and ends in 3 zeros.

E.g. 195 to the nearest 10 is 200 and is 5 away from the original number.

Task B: Degrees of accuracy

1) Match each statement to the most appropriate **degree of accuracy**:

The holiday could be nearest hundred or thousand.

The number of sweets in a bag

Nearest million

The population of the UK

Nearest hundred

The number of people at a national sporting event

Nearest ten

Cost of a holiday

Nearest thousand

2) Match each statement to the most appropriate **degree of accuracy**:

Length of a pencil in centimetres

Nearest integer

The length of a piece of ribbon in metres

1 decimal place

Width of a doorway in metres

2 decimal places

Capacity of a mug in millilitres

3 decimal places

3) Answer the following, make sure that you carefully consider the appropriate **degree of accuracy**.

- a) Izzy collects 100 eggs. Boxes hold 6 eggs. How many boxes does Izzy need for all her eggs?

$100 \div 6 = 16.667$, Izzy needs 17 boxes.

- b) Jun decides to share his bag of 200 sweets with Sofia and Sam. They each get the same number of sweets. How many do they each get?

$200 \div 3 = 66.667$, they each get 66 sweets.

- c) Aisha has baked 150 cookies for the school fete. She is putting them into boxes of 4. How many boxes can Aisha make?

$150 \div 4 = 37.5$, Aisha can make 37 boxes of cookies.

- d) A group of 56 scouts are going on a camping trip. Each tent sleeps 6. How many tents are needed for the scouts?

$56 \div 6 = 9.333$, they will need 10 tents.